25900511 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEMAT200 - ACTUARIAL MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

{Short Answer Questions}

Answer any **5** questions

Each question carries **2** marks

1. State the mean and variance in terms of moments [A] / [CO1]
2. State the definition of a joint distribution of three random variables X, Y, Z [A] / [CO2]
3. What is meant by geometric Brownian motion in stochastic processes? [U] / [CO3]
4. Define the first hitting time T_a for Brownian motion [U] / [CO3]
5. Define the present value of the benefits on a discrete time multi-state insurance contract. [K] / [CO4]
6. Consider a two step path $\pi = (i, k, j)$. Find the probability $p_{ij}^{\pi}(s, t)$. [U] / [CO4]
7. Difference between European call and put in BSM. [U] / [CO5]
8. What is the payoff of a European call option? [U] / [CO5]

[2x5 = 10]

Part B

{Short Essay Questions}

Answer any **5** questions

Each question carries **6** marks

9.	Define the probability distribution function of a Random variable.	[A] / [CO1, CO2]
10.	Distinguish between the absorbing and recurrent states of a stationary Markov chain.	[A] / [CO2]
11.	What is a counting process? Describe its properties and give an example from a real-life application.	[U] / [CO3]
12.	Given two time point s and t, what is the conditional distribution of Brownian motoin B_t given that $B_s = b$	[A] / [CO3]
13.	Give the recursion and differential equations for multi-state reserves. Explain the terms.	[K] / [CO4]
14.	Let i be a state such that $\mu_{ij}(t) = 0$ for all $j \neq i$ and all t. Then prove that for s < t $p_{ii}(s,t) = e^{\int_s^t \mu_{ii}(r)dr}$ and $p_{ji}(s,t) = 0$ for $j \neq i$	[U] / [CO4]
15.	Which are the steps to check if a market is arbitrage-free.	[U] / [CO5]
16.	Derive the expression for a two-step binomial model of asset prices	[An] / [CO4]

[6x5 = 30]

Part C

{Essay Questions}

Answer any **3** questions

Each question carries **10** marks

17.	Explain the “Service Table” model of Mortality used by Pension Actuaries in connection with Employer Sponsored Pension Plans.	[A] / [CO1, CO2]
18.	Explain the Stochastic Process, “Gambler’s fortune”, without Markov property	[A] / [CO2]
19.	1.Define a Brownian motion and standard Brownian motion process with variance parameter σ^2 2.Differentiate Poisson process and Brownian motion process.	[A] / [CO3]
20.	1.Define the waiting-time process associated with a counting process $N(t)$ 2.For a Poisson process with rateλ prove that the inter-arrival times W_n are i.i.d. exponential random variables with parameter λ.	[A] / [CO3]
21.	(i) Suppose that the force of interest and all forces of transition are constant. Find a formula for $\overline{A}_\pi$, the acturial present value of the insurance based on the path π. (ii) In the common shock model calculate the probability that at time n, the forst type of failure will have occured but not the second.	[U] / [CO4]
22.	In a one-period model with $S_0 = 50, u = 1.5, d = 0.5, r = 0.1$, (a) check arbitrage, (b) price a European put with strike 50.	[A] / [CO5]

[10x3 = 30]